

Math 871 Midterm 1 solutions

Problem 1. Define Hausdorff space.

Solution. A topological space X is Hausdorff if for all distinct points x and y in X , there exist open sets U and V such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

Problem 2. For each of the following, give an example **without proof**:

a) A topological space $(X, \mathcal{T})$ and a nonempty subset $U \subsetneq X$ that is both open and closed.

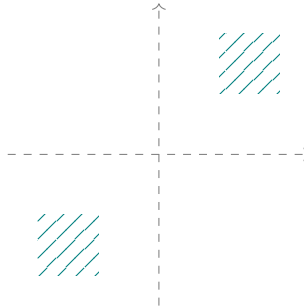
Solution. Consider $X = \mathbb{N}$ under the discrete topology and $U = \{0\}$.

b) Topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ and an open set W in $X \times Y$ with the product topology that is not of the form $U \times V$ with $U \in \mathcal{T}_X$ and $V \in \mathcal{T}_Y$.

Solution. Take $X = Y = \mathbb{R}$ under the standard topology and

$$W =]-2, -1[\times]-2, -1[\cup]1, 2[\times]1, 2[.$$

Not needed, but here's a nice picture:



Alternative Solution. Take $X = Y = \mathbb{R}$ with the standard topology and W to be an open ball in $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$; eg, take W to be the ball of radius 1 centered at $(0, 0)$.

Problem 3.

a) Show that if $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X , then $\mathcal{T}_1 \cap \mathcal{T}_2$ is also a topology on X .

Solution. Since $\mathcal{T}_1$ is a topology, $\emptyset, X \in \mathcal{T}_1$. Since $\mathcal{T}_2$ is a topology, $\emptyset, X \in \mathcal{T}_2$. Thus $\emptyset, X \in \mathcal{T}_1 \cap \mathcal{T}_2$.

Consider a collection $\{U_\alpha\}_{\alpha \in I}$ of elements of $\mathcal{T}_1 \cap \mathcal{T}_2$. Since $\mathcal{T}_1$ is a topology,

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}_1.$$

Since $\mathcal{T}_2$ is a topology,

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}_2.$$

Therefore,

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}_1 \cap \mathcal{T}_2.$$

Finally, suppose that $U, V \in \mathcal{T}_1 \cap \mathcal{T}_2$. Since $\mathcal{T}_1$ is a topology, $U \cap V \in \mathcal{T}_1$. Since $\mathcal{T}_2$ is a topology, $U \cap V \in \mathcal{T}_2$. Therefore, $U \cap V \in \mathcal{T}_1 \cap \mathcal{T}_2$.

- b) Consider the lower limit topology $\mathcal{T}_\ell$ on $\mathbb{R}$, generated by the basis $\{[a, b) \mid a < b \in \mathbb{R}\}$, and the upper limit topology $\mathcal{T}_u$, generated by the basis $\{(a, b] \mid a < b \in \mathbb{R}\}$. Prove that $\mathcal{T}_\ell \cap \mathcal{T}_u = \mathcal{T}_{\text{std}}$, the standard topology on $\mathbb{R}$.

Solution. Fix real numbers $a < b$. We have

$$(a, b) = \bigcup_{n \geq 1} \left[a + \frac{1}{n}, b \right) = \bigcup_{n \geq 1} \left(a, b - \frac{1}{n} \right].$$

Thus (a, b) is a union of basis elements in $\mathcal{T}_1$, and thus $(a, b) \in \mathcal{T}_1$, and (a, b) is a union of basis elements in $\mathcal{T}_2$, and thus $(a, b) \in \mathcal{T}_2$. Since $\{(a, b) \mid a < b \in \mathbb{R}\}$ forms a basis for the standard topology, we have proved that $\mathcal{T}_{\text{std}} \subseteq \mathcal{T}_\ell \cap \mathcal{T}_u$.

Now consider any $U \in \mathcal{T}_1 \cap \mathcal{T}_2$. and any $x \in U$. Since $U \in \mathcal{T}_1$ there exist $a < b$ such that $x \in [a, b) \subseteq U$. Since $U \in \mathcal{T}_2$ there exist $c < d$ such that $x \in (c, d] \subseteq U$. Since $a < b$ and $c < d$,

$$m = \min(a, c) < \max(b, d) = M.$$

Moreover,

$$m \leq c < x < b \leq M \implies m < M.$$

We conclude that $x \in (c, d) \subseteq U$. Since this holds for all $x \in U$, we have $U \in \mathcal{T}_{\text{std}}$, and therefore $\mathcal{T}_\ell \cap \mathcal{T}_u \subseteq \mathcal{T}_{\text{std}}$.

Problem 4. Let A and B be subsets of a topological space X .

- a) Prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Proof. ($\supseteq$) If $U \subseteq V$, then $\overline{U} \subseteq \overline{V}$. Since $A \subseteq A \cup B$, then $\overline{A} \subseteq \overline{A \cup B}$. Similarly, $\overline{B} \subseteq \overline{A \cup B}$. Thus $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$.

($\subseteq$) By (a), $A \subseteq \overline{A}$ and $B \subseteq \overline{B}$, so

$$A \cup B \subseteq \overline{A} \cup \overline{B}.$$

The finite union of closed sets is closed, so $\overline{A \cup B}$ is closed. Any closed set that contains $A \cup B$ also contains its closure, and thus

$$\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}. \quad \square$$

- b) Give an example (with proof) where $\overline{A \cap B} \neq \overline{A} \cap \overline{B}$.

Proof. Consider $\mathbb{R}$ with the standard topology, $A = (0, 1)$, and $B = (1, 2)$. Then $A \cap B = \emptyset$, so $\overline{A \cap B} = \overline{\emptyset} = \emptyset$, while $\overline{A} = [0, 1]$, $\overline{B} = [1, 2]$, and $\overline{A} \cap \overline{B} = \{1\} \neq \emptyset$. $\square$

c) Show that if A is closed and B is open in X , then $A \setminus B$ is closed.

Solution. Since B is open, then $X \setminus B$ is closed, and thus

$$A \setminus B = \{x \in A \mid x \notin B\} = \{x \in X \mid x \in A \text{ and } x \in X \setminus B\} = A \cap (X \setminus B)$$

is an intersection of closed sets, and thus closed.

Alternative proof. Since A is closed, $X \setminus A$ is open, and thus

$$X \setminus (A \setminus B) = X \setminus A \cup B$$

is a union of two open sets, and thus open.

Choose one of the following two problems.

Problem 5. Let $(X, \mathcal{T})$ be a topological space. A subset $A \subseteq X$ is **nowhere dense** if its closure $\overline{A}$ has empty interior. Show that if U is an open set, then $\overline{U} \setminus U$ is nowhere dense.

Solution. First, note that A has empty interior if and only there are no nonempty open sets V such that $V \subseteq A$.

Let U be open. By 4 c), $\overline{U} \setminus U$ is closed, so its closure is itself. Thus we want to show that there are no nonempty open sets V contained in $\overline{U} \setminus U$. Let $V \subseteq \overline{U} \setminus U$ be an open set. If $x \in V$, then $x \in V \subseteq \overline{U}$ and thus any open neighborhood of x intersects U . In particular, $V \cap U = \emptyset$. However, $V \subseteq \overline{U} \setminus U \subseteq X \setminus U$, so $V \cap U = \emptyset$. We conclude that no such x can exist, and thus V is empty. Therefore, there are no nonempty open sets contained in $\overline{U} \setminus U$, and thus $\overline{U} \setminus U$ is nowhere dense.

Problem 6. A topological space $(X, \mathcal{T})$ is called **clumped** if for any two nonempty open sets U and V intersect nontrivially, meaning $U \cap V \neq \emptyset$. Show that if X and Y are clumped, then $X \times Y$ is clumped in the product topology.

Solution. Recall that

$$\mathcal{B} = \{U \times V \mid U \text{ open in } X, V \text{ open in } Y\}$$

forms a basis for the product topology on $X \times Y$. Suppose that X and Y are clumped, and let A and B be two open sets in $X \times Y$. Since $\mathcal{B}$ is a basis for the topology on $X \times Y$, and A is nonempty, A contains an open set in $\mathcal{B}$, and thus of the form $U_1 \times V_1$ with U_1 open in X and V_1 open in Y . Similarly, B contains an open set of the form $U_2 \times V_2$ with U_2 open in X and V_2 open in Y . Since X is clumped, $U_1 \cap U_2 \neq \emptyset$. Since Y is clumped, $V_1 \cap V_2 \neq \emptyset$. Let $x \in U_1 \cap U_2$ and $y \in V_1 \cap V_2$. Then

$$(x, y) \in U_1 \times V_1 \quad \text{and} \quad (x, y) \in U_2 \times V_2 \implies (x, y) \in U_1 \times V_1 \cap U_2 \times V_2.$$

Therefore, $U_1 \times V_1 \cap U_2 \times V_2 \neq \emptyset$. Since $U_1 \times V_1 \subseteq A$ and $U_2 \times V_2 \subseteq B$, we conclude that $A \cap B \neq \emptyset$.